

- (c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

- (i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

- (ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

							parity bit ↓
1	0	1	1	0	1	1	1
0	1	1	1	0	0	0	0
0	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer.

[1]

- (d) A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

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File size megabytes

[2]

